

SQL DEMOGRAPHIC ANALYSIS • 2008–2014

California Education & Income Analysis

Seven years of population data across age, gender, education, and personal income — explored with joins, window functions, CTEs, and subqueries.

1,060 records

7 years

5 education levels

8 income bands

What the data covers, and what we set out to find

The dataset tracks California's population by Age group, Gender, Educational Attainment, and Personal Income band, for every year from 2008 to 2014. The goal was to move beyond a single table of numbers and use SQL (joins, window functions, CTEs, and subqueries) to uncover how education and income shifted across the recession-recovery period.

1,060

Total records

7

Years (2008–14)

3

Age groups

2

Genders

5

Education levels

8

Income bands

Guiding questions

- How did total population, and the male–female balance, evolve 2008–14?
- Which education levels and income bands grew the fastest — and which shrank?
- Which demographic segments gained the most people, and which declined?

SQL techniques used to answer 28 analytical questions

1

Self Joins

Paired each segment's 2008 row against its 2014 row to measure growth directly, combo by combo.

2

Window Functions

RANK, DENSE_RANK, ROW_NUMBER and LAG to rank categories and track year-over-year change.

3

CTEs

Layered WITH clauses to build yearly aggregates, then classify, compare, and filter on top of them.

4

Subqueries

Correlated and nested subqueries to find above-average, top, and second-highest segments.

- 1 Compare 2008 and 2014 population for every combination of Age, Gender, Educational Attainment and Personal Income. Show population in both years and the change.

SELECT

```
a.age,  
a.gender,  
a.educational_attainment,  
a.personal_income,  
a.population_count AS population_2008,  
b.population_count AS population_2014,  
b.population_count - a.population_count AS population_change  
FROM ca_education_income a
```

age	gender	educational_attainment	personal_income	population_2008	population_2014	population_change
00 to 17	Male	Children under 15	No Income	0	0	0
00 to 17	Male	No high school diploma	No Income	650889	610313	-40576
00 to 17	Male	No high school diploma	\$5,000 to \$9,999	30152	26380	-3772
00 to 17	Male	No high school diploma	\$10,000 to \$14,999	7092	11857	4765
00 to 17	Male	No high school diploma	\$15,000 to \$24,999	3974	3920	-54
00 to 17	Male	High school or equivalent	No Income	0	15741	15741
00 to 17	Male	Some college, less than 4-yr degree	\$15,000 to \$24,999	3465	1664	-1801
00 to 17	Female	Children under 15	No Income	0	0	0
00 to 17	Female	No high school diploma	No Income	635274	639596	4322

- 2 Find demographic segments whose population increased by more than 10,000 people between 2008 and 2014. Use a SELF JOIN.

SELECT

```
a.age,  
a.gender,  
a.educational_attainment,  
a.personal_income,  
a.population_count AS population_2008,  
b.population_count AS population_2014,  
b.population_count - a.population_count AS population_increase  
FROM ca_education_income a  
JOIN ca_education_income b
```

age	gender	educational_attainment	personal_income	population_2008	population_2014	population_increase
00 to 17	Male	High school or equivalent	No Income	0	15741	15741
18 to 64	Male	No high school diploma	No Income	261172	376783	115611
18 to 64	Male	No high school diploma	\$75,000 and over	44291	54734	10443
18 to 64	Male	High school or equivalent	No Income	296555	405204	108649
18 to 64	Male	High school or equivalent	\$5,000 to \$9,999	187833	238754	50921
18 to 64	Male	High school or equivalent	\$15,000 to \$24,999	501296	520580	19284
18 to 64	Male	High school or equivalent	\$25,000 to \$34,999	433340	467906	34566
18 to 64	Male	High school or equivalent	\$75,000 and over	175104	227515	52411

- 3 Compare the population for each Gender + Educational Attainment combination between 2008 and 2014, aggregating across Age and Personal Income.

```
SELECT
    a.gender,
    a.educational_attainment,
    SUM(a.population_count) AS population_2008,
    SUM(b.population_count) AS population_2014,
    SUM(b.population_count) - SUM(a.population_count) AS population_change
FROM ca_education_income a
JOIN ca_education_income b
    ON a.age = b.age
```

gender	educational_attainment	population_2008	population_2014	population_change
Male	Children under 15	0	0	0
Male	No high school diploma	3073168	2811001	-262167
Male	High school or equivalent	3081357	3294850	213493
Male	Some college, less than 4-yr degree	3487582	3685424	197842
Female	Children under 15	0	0	0
Female	No high school diploma	2882597	2768905	-113692
Female	High school or equivalent	2876358	3016615	140257
Female	Some college, less than 4-yr degree	3739433	4093093	353660

- 4 Compare Male and Female population for each Educational Attainment and Personal Income combination in 2014. Show the gender difference.

```
SELECT
    a.educational_attainment,
    a.personal_income,
    a.population_count AS male_population,
    b.population_count AS female_population,
    a.population_count - b.population_count AS gender_difference
FROM ca_education_income a
JOIN ca_education_income b
    ON a.educational_attainment = b.educational_attainment
    AND a.personal_income = b.personal_income
```

educational_attainment	personal_income	male_population	female_population	gender_difference
Children under 15	No Income	0	0	0
No high school diploma	No Income	31263	639596	-608333
No high school diploma	No Income	376783	639596	-262813
No high school diploma	No Income	610313	639596	-29283
No high school diploma	\$5,000 to \$9,999	40404	15795	24609
No high school diploma	\$5,000 to \$9,999	158572	15795	142777
No high school diploma	\$5,000 to \$9,999	26380	15795	10585
No high school diploma	\$10,000 to \$14,999	81810	13768	68042

5

Compare 2013 and 2014 population for every Age + Gender segment, aggregating across Education and Income. Find segments that declined.

SELECT

```

a.age,
a.gender,
SUM(a.population_count) AS population_2013,
SUM(b.population_count) AS population_2014,
SUM(b.population_count) - SUM(a.population_count) AS population_change
FROM ca_education_income a
JOIN ca_education_income b
  ON a.age = b.age
  AND a.gender = b.gender
  AND a.educational_attainment = b.educational_attainment
  AND a.personal_income = b.personal_income

```

age	gender	population_2013	population_2014	population_change
00 to 17	Male	688204	670259	-17945
00 to 17	Female	707295	675989	-31306

6

For each Personal Income band, compare total population in 2008 versus 2014, aggregating across Age, Gender and Education.

SELECT

```

a.personal_income,
SUM(a.population_count) AS population_2008,
SUM(b.population_count) AS population_2014,
SUM(b.population_count) - SUM(a.population_count) AS population_change
FROM ca_education_income a
JOIN ca_education_income b
  ON a.age = b.age
  AND a.gender = b.gender
  AND a.educational_attainment = b.educational_attainment

```

personal_income	population_2008	population_2014	population_change
No Income	4331910	5192554	860644
\$5,000 to \$9,999	2317874	2405313	87439
\$10,000 to \$14,999	2652090	2633938	-18152
\$15,000 to \$24,999	4036167	4252400	216233
\$25,000 to \$34,999	3003788	3179308	175520
\$35,000 to \$49,999	3319746	3278595	-41151
\$50,000 to \$74,999	3354245	3136540	-217705
\$75,000 and over	3497639	4116938	619299

7 Rank Educational Attainment categories by total population within each year using RANK().

```
SELECT
    SUBSTRING_INDEX(year, '/', -1) AS year,
    educational_attainment,
    SUM(population_count) AS total_population,
    RANK() OVER (
        PARTITION BY SUBSTRING_INDEX(year, '/', -1)
        ORDER BY SUM(population_count) DESC
    ) AS population_rank
FROM ca_education_income
```

year	educational_attainment	total_population	population_rank
2008	Bachelor's degree or higher	7372964	1
2008	Some college, less than 4-yr degree	7236983	2
2008	No high school diploma	5964588	3
2008	High school or equivalent	5957715	4
2008	Children under 15	0	5
2009	Bachelor's degree or higher	7436401	1
2009	Some college, less than 4-yr degree	7255063	2
2009	No high school diploma	5902137	3

8 Rank Personal Income bands by population within each year using DENSE_RANK().

```
SELECT
    SUBSTRING_INDEX(year, '/', -1) AS year,
    personal_income,
    SUM(population_count) AS total_population,
    DENSE_RANK() OVER (
        PARTITION BY SUBSTRING_INDEX(year, '/', -1)
        ORDER BY SUM(population_count) DESC
    ) AS population_rank
FROM ca_education_income
```

year	personal_income	total_population	population_rank
2008	No Income	4340574	1
2008	\$15,000 to \$24,999	4038176	2
2008	\$75,000 and over	3499620	3
2008	\$50,000 to \$74,999	3354245	4
2008	\$35,000 to \$49,999	3321973	5
2008	\$25,000 to \$34,999	3006394	6
2008	\$10,000 to \$14,999	2653394	7
2008	\$5,000 to \$9,999	2317874	8

9

Find the top 3 Educational Attainment categories by population for every year using ROW_NUMBER().

```
SELECT
    year,
    educational_attainment,
    total_population,
    population_rank
FROM (
    SELECT
        SUBSTRING_INDEX(year, '/', -1) AS year,
        educational_attainment,
        SUM(population_count) AS total_population,
        ROW_NUMBER() OVER (
```

year	educational_attainment	total_population	population_rank
2008	Bachelor's degree or higher	7372964	1
2008	Some college, less than 4-yr degree	7236983	2
2008	No high school diploma	5964588	3
2009	Bachelor's degree or higher	7436401	1
2009	Some college, less than 4-yr degree	7255063	2
2009	No high school diploma	5902137	3
2010	Bachelor's degree or higher	7648905	1
2010	Some college, less than 4-yr degree	7186726	2

10

Calculate previous-year population for every Gender +Educational Attainment combination using LAG(), and calculate the population change.

```
SELECT
    year,
    gender,
    educational_attainment,
    total_population,
    LAG(total_population) OVER (
        PARTITION BY gender, educational_attainment
        ORDER BY year
    ) AS previous_population,
```

year	gender	educational_attainment	total_population	previous_population	population_change
2008	Female	Bachelor's degree or higher	3630986	NULL	NULL
2009	Female	Bachelor's degree or higher	3587347	3630986	-43639
2010	Female	Bachelor's degree or higher	3753679	3587347	166332
2011	Female	Bachelor's degree or higher	3662573	3753679	-91106
2012	Female	Bachelor's degree or higher	3800850	3662573	138277
2013	Female	Bachelor's degree or higher	4011603	3800850	210753
2014	Female	Bachelor's degree or higher	4201713	4011603	190110
2008	Female	Children under 15	0	NULL	NULL

- 11 Calculate year-over-year population growth percentage for every Age + Gender combination using LAG().

```
SELECT
    year,
    age,
    gender,
    total_population,
    LAG(total_population) OVER (
        PARTITION BY age, gender
        ORDER BY year
    ) AS previous_population,
    ROUND(
        (total_population - LAG(total_population) OVER (
            PARTITION BY age, gender
            ORDER BY year
        )) / LAG(total_population) OVER (
```

year	age	gender	total_population	previous_population	growth_percentage
2008	00 to 17	Female	698888	NULL	NULL
2009	00 to 17	Female	712227	698888	1.91
2010	00 to 17	Female	663685	712227	-6.82
2011	00 to 17	Female	695767	663685	4.83
2012	00 to 17	Female	694577	695767	-0.17
2013	00 to 17	Female	715607	694577	3.03
2014	00 to 17	Female	685916	715607	-4.15
2008	00 to 17	Male	710373	NULL	NULL

- 12 Calculate cumulative population for each Educational Attainment category from 2008 through 2014.

```
SELECT
    year,
    educational_attainment,
    total_population,
    SUM(total_population) OVER (
        PARTITION BY educational_attainment
        ORDER BY year
    ) AS cumulative_population
FROM (
    SELECT
        SUBSTRING_INDEX(year, '/', -1) AS year,
```

year	educational_attainment	total_population	cumulative_population
2008	Bachelor's degree or higher	7372964	7372964
2009	Bachelor's degree or higher	7436401	14809365
2010	Bachelor's degree or higher	7648905	22458270
2011	Bachelor's degree or higher	7564167	30022437
2012	Bachelor's degree or higher	7765833	37788270
2013	Bachelor's degree or higher	8301237	46089507
2014	Bachelor's degree or higher	8528169	54617676
2008	Children under 15	0	0

13 Calculate each Personal Income band's percentage share of total population within each year.

```
SELECT
    year,
    personal_income,
    total_population,
    ROUND(
        total_population /
        SUM(total_population) OVER (PARTITION BY year) * 100,
        2
    ) AS percentage_share
FROM (
    SELECT
        SUBSTRING_INDEX(year, '/', -1) AS year,
```

year	personal_income	total_population	percentage_share
2008	No Income	4340574	16.36
2008	\$5,000 to \$9,999	2317874	8.74
2008	\$10,000 to \$14,999	2653394	10.00
2008	\$15,000 to \$24,999	4038176	15.22
2008	\$25,000 to \$34,999	3006394	11.33
2008	\$35,000 to \$49,999	3321973	12.52
2008	\$50,000 to \$74,999	3354245	12.64
2008	\$75,000 and over	3499620	13.19

14 For every year, identify the most populated Gender + Age segment using ROW_NUMBER().

```
SELECT
    year,
    gender,
    age,
    total_population
FROM (
    SELECT
        SUBSTRING_INDEX(year, '/', -1) AS year,
        gender,
        age,
        SUM(population_count) AS total_population,
        ROW_NUMBER() OVER (
```

year	gender	age	total_population
2008	Male	18 to 64	11072116
2009	Male	18 to 64	11040468
2010	Male	18 to 64	10954078
2011	Male	18 to 64	11115121
2012	Male	18 to 64	11254245
2013	Male	18 to 64	11323361
2014	Male	18 to 64	11382669

- 15 Find the year in which each Educational Attainment category had its highest population.

```
SELECT
    year,
    educational_attainment,
    total_population
FROM (
    SELECT
        SUBSTRING_INDEX(year, '/', -1) AS year,
        educational_attainment,
        SUM(population_count) AS total_population,
        ROW_NUMBER() OVER (
            PARTITION BY educational_attainment
            ORDER BY SUM(population_count) DESC
```

year	educational_attainment	total_population
2014	Bachelor's degree or higher	8528169
2010	Children under 15	0
2012	High school or equivalent	6372967
2012	No high school diploma	6002290
2013	Some college, less than 4-yr degree	7850041

- 16 Compare each year's total population with the previous year using LAG(), and identify the largest annual increase.

```
SELECT
    year,
    total_population,
    previous_population,
    total_population - previous_population AS population_increase
FROM (
    SELECT
        year,
        total_population,
        LAG(total_population) OVER (
            ORDER BY year
        ) AS previous_population
```

year	total_population	previous_population	population_increase
2011	27160088	26469031	691057

17

Using CTEs, calculate total population by year and classify each year as High Population or Low Population based on the overall average.

```
WITH yearly_population AS (
  SELECT
    SUBSTRING_INDEX(year, '/', -1) AS year,
    SUM(population_count) AS total_population
  FROM ca_education_income
  GROUP BY SUBSTRING_INDEX(year, '/', -1)
),
average_population AS (
  SELECT
    AVG(total_population) AS overall_average
  FROM yearly_population
)
```

year	total_population	population_category
2008	26532250	Low Population
2009	26442817	Low Population
2010	26469031	Low Population
2011	27160088	Low Population
2012	27641508	High Population
2013	27905466	High Population
2014	28215807	High Population

18

Using CTEs, calculate Educational Attainment population in 2008 and 2014 and return the top 5 categories by absolute change.

```
WITH education_population AS (
  SELECT
    educational_attainment,
    SUBSTRING_INDEX(year, '/', -1) AS year,
    SUM(population_count) AS total_population
  FROM ca_education_income
  WHERE SUBSTRING_INDEX(year, '/', -1) IN ('2008', '2014')
  GROUP BY
    educational_attainment,
    SUBSTRING_INDEX(year, '/', -1)
),
education_change AS (
```

educational_attainment	population_2008	population_2014	population_change	absolute_change
Bachelor's degree or higher	7372964	8528169	1155205	1155205
Some college, less than 4-yr degree	7236983	7781293	544310	544310
No high school diploma	5964588	5583171	-381417	381417
High school or equivalent	5957715	6323174	365459	365459
Children under 15	0	0	0	0

19

Using CTEs, calculate population by Gender and Year and find which gender had the larger population in each year.

```
WITH gender_population AS (
    SELECT
        SUBSTRING_INDEX(year, '/', -1) AS year,
        gender,
        SUM(population_count) AS total_population
    FROM ca_education_income
    GROUP BY
        SUBSTRING_INDEX(year, '/', -1),
        gender
)
```

year	male_population	female_population	larger_gender
2008	13398886	13133364	Male
2009	13429050	13013767	Male
2010	13356399	13112632	Male
2011	13720944	13439144	Male
2012	13929862	13711646	Male
2013	14036096	13869370	Male
2014	14124134	14091673	Male

20

Using CTEs, calculate average annual population for each Age group and identify the Age group with the highest average.

```
WITH age_population AS (
    SELECT
        age,
        SUM(population_count) AS total_population
    FROM ca_education_income
    GROUP BY age
),
age_average AS (
    SELECT
        age,
        total_population / 7 AS average_population
    FROM age_population
)
```

year	male_population	female_population	larger_gender
2008	13398886	13133364	Male
2009	13429050	13013767	Male
2010	13356399	13112632	Male
2011	13720944	13439144	Male
2012	13929862	13711646	Male
2013	14036096	13869370	Male
2014	14124134	14091673	Male

21

Using CTEs, identify Educational Attainment categories whose population increased in at least 4 of the 6 year-to-year transitions.

```
WITH education_population AS (
    SELECT
        SUBSTRING_INDEX(year, '/', -1) AS year,
        educational_attainment,
        SUM(population_count) AS total_population
    FROM ca_education_income
    GROUP BY
        SUBSTRING_INDEX(year, '/', -1),
        educational_attainment
),
yearly_change AS (
    SELECT
```

educational_attainment	increase_years
Bachelor's degree or higher	5
High school or equivalent	4
Some college, less than 4-yr degree	4

22

Using CTEs, calculate the average yearly population share of each Personal Income band and find bands whose average share exceeds 10%.

```
WITH income_population AS (
    SELECT
        SUBSTRING_INDEX(year, '/', -1) AS year,
        personal_income,
        SUM(population_count) AS income_population
    FROM ca_education_income
    GROUP BY
        SUBSTRING_INDEX(year, '/', -1),
        personal_income
),
yearly_share AS (
    SELECT
```

personal_income	average_yearly_share
No Income	17.40
\$15,000 to \$24,999	15.47
\$75,000 and over	13.66
\$35,000 to \$49,999	11.77
\$50,000 to \$74,999	11.74
\$25,000 to \$34,999	11.16

23

Find Educational Attainment categories whose 2014 population was greater than the average 2014 population across all education categories. Use a subquery.

```
SELECT
    educational_attainment,
    SUM(population_count) AS population_2014
FROM ca_education_income
WHERE SUBSTRING_INDEX(year, '/', -1) = '2014'
GROUP BY educational_attainment
HAVING SUM(population_count) > (
    SELECT AVG(total_population)
    FROM (
        SELECT
            educational_attainment,
            SUM(population_count) AS total_population
        FROM ca_education_income
```

educational_attainment	population_2014
Bachelor's degree or higher	8528169
Some college, less than 4-yr degree	7781293
High school or equivalent	6323174

24

Find Personal Income bands whose 2014 population was greater than the average income-band population in 2014. Use a subquery.

```
SELECT
    personal_income,
    SUM(population_count) AS population_2014
FROM ca_education_income
WHERE SUBSTRING_INDEX(year, '/', -1) = '2014'
GROUP BY personal_income
HAVING SUM(population_count) > (
    SELECT AVG(total_population)
```

personal_income	population_2014
No Income	5192554
\$15,000 to \$24,999	4260763
\$75,000 and over	4116938

25

Find the most populated Age + Gender combination in 2014 using a subquery.

```
SELECT
    age,
    gender,
    SUM(population_count) AS population_2014
FROM ca_education_income
WHERE SUBSTRING_INDEX(year, '/', -1) = '2014'
GROUP BY age, gender
HAVING SUM(population_count) = (
    SELECT MAX(total_population)
    FROM (
        SELECT
```

age	gender	population_2014
18 to 64	Male	11382669

26

Find the second-highest Educational Attainment category by population in 2014 using subqueries.

```
SELECT
    educational_attainment,
    SUM(population_count) AS population_2014
FROM ca_education_income
WHERE SUBSTRING_INDEX(year, '/', -1) = '2014'
GROUP BY educational_attainment
HAVING SUM(population_count) = (
    SELECT MAX(total_population)
    FROM (
        SELECT
            educational_attainment,
            SUM(population_count) AS total_population
```

educational_attainment	population_2014
Some college, less than 4-yr degree	7781293

27

Find Age + Gender combinations whose average annual population from 2008-2014 is above the overall average annual population across all Age + Gender combinations.

```
SELECT
    age,
    gender,
    AVG(total_population) AS average_annual_population
FROM (
    SELECT
        age,
        gender,
        SUBSTRING_INDEX(year, '/', -1) AS year,
        SUM(population_count) AS total_population
    FROM ca_education_income
    GROUP BY
        age,
        gender,
```

age	gender	average_annual_population
18 to 64	Male	11163151.1429
18 to 64	Female	10523548.1429

28

Find Educational Attainment categories whose 2014 population was higher than their own average population across 2008-2014.

```
SELECT
    educational_attainment,
    population_2014,
    average_population
FROM (
    SELECT
        educational_attainment,
        SUM(CASE
            WHEN SUBSTRING_INDEX(year, '/', -1) = '2014'
            THEN population_count
            ELSE 0
        END) AS population_2014,
```

educational_attainment	population_2014	average_population
Bachelor's degree or higher	8528169	7802525.1429
High school or equivalent	6323174	6134703.7143
Some college, less than 4-yr degree	7781293	7448213.4286

California Education & Income, 2008–2014

What seven years of population data reveal, by age, gender, education, and personal income.

+6.3%

Steady population growth

Total population rose from 26.5M to 28.2M — dipping only in 2009, then growing every year through 2014.

32K

Gender gap nearly closed

The male–female population gap shrank from 265K in 2008 to just 32K by 2014.

+1.16M

Education paid off

Bachelor's degree holders grew the most of any group; the no-diploma population was the only category to shrink (–381K).

18.4%

More Californians report no income

The “No Income” bracket's share of the population rose from 16.4% (2008) to 18.4% (2014).

+20%

Population is aging

The 65–80+ age group grew fastest of all three age bands; the under-18 population slightly declined.

+172K

Top-gaining segment

Working-age (18–64) adults with a bachelor's degree earning \$75,000+ saw the single largest population gains, in both genders.

Highly-educated, high-earning adults gained the most people

Age	Gender	Education	Income	2008	2014	Change
18 to 64	Female	Bachelor's degree or higher	\$75,000 and over	781,211	953,282	+172,071
18 to 64	Male	Bachelor's degree or higher	\$75,000 and over	1,456,566	1,628,605	+172,039
18 to 64	Female	High school or equivalent	No Income	528,876	670,294	+141,418
18 to 64	Male	Some college, <4-yr degree	\$15,000–\$24,999	416,638	556,174	+139,536
18 to 64	Female	Bachelor's degree or higher	No Income	224,757	359,264	+134,507

Reading the pattern

Four of the top five gaining segments are working-age adults (18–64) with a Bachelor's degree or higher. The two largest gains — women and men earning \$75,000+ with a bachelor's degree — grew by almost identical amounts, suggesting the recovery years disproportionately benefited highly-educated, high-earning Californians.



CONCLUSION

Actionable Socioeconomic Impact

This project successfully connects demographic population counts, educational attainment milestones, and personal income distribution into a unified, high-performance MySQL 8.0 analytical framework ready for executive deployment.



Thank you!

California Educational Attainment & Personal Income (2008-2014)— SQL Project

BY- MD SHAH FAISHAL